Al Imam Mohammad Ibn Saud Islamic University
College of Computer and Information Sciences

Computer Science Department

Course Title:	Design and Analysis of Algorithms
Course Code:	CS215
Course Instructors:	Prof. Hachemi Bennaceur, Mrs Ghada Farid, Mrs Hayat Almosalam, Mrs Mai Al-Ammar, Dr. Miled Tezeghdanti, Dr. Yassine Daadaa
Exam:	Final Exam
Semester:	First Semester
Date:	20/12/2015 – 09/03/1437
Duration:	120 minutes
Marks:	40

Privilege: ☐ Open Book ☐ Open Notes ☐ Calculator Permitted ☐ Laptop Permitted

Student Name (in English):	
Student ID:	
Section No.:	

Instructions:

1. Answer **5** questions; there are **5** questions in **7** pages.
2. Write your name on each page of the exam paper.
3. Write your answers directly on the question sheets. Use the ends of the question pages for rough work or if you need extra space for your answer.
4. If information appears to be missing from a question, make a reasonable assumption, state your assumption, and proceed.
5. **No** questions will be answered by the invigilator(s) during the exam period.

Official Use Only		
Question	**Student Marks**	**Question Marks**
1		13
2		10
3		6
4		11
Total		**40**

Student Name (in English): _____ Student ID: _____

Question 1: To be answered in (20) Minutes [ ] / 13 Marks

a. Choose from column B the *most* appropriate number for each sentence in column A. [4 Marks]

(0.5 mark each)

	Column A		Column B
8	It chooses the Locally Optimal Choice hoping that it would give a Globally Optimal Solution.	**1**	Prim's algorithm
7	Very general problem-solving technique that consists of systematically enumerating all possible solutions and tests them	**2**	Matrix chain multiplication
6	It permits negative-weight edges in calculating the shortest path.	**3**	Divide and conquer
4	It improves performance because partial results are never calculated twice.	**4**	Memoization
1	An algorithm that is used to compute minimal spanning tree.	**5**	Optimization problem
5	Trying to find the best solution from all feasible solutions.	**6**	Bellman-Ford Algorithm
3	Merge sort is an example of this technique	**7**	Brute force technique
2	An optimization problem that can be solved using dynamic programming	**8**	Greedy approach

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b. Complete the following table. [4 Marks]

(0.5 mark each)

Algorithm	**Worst Case**
Selection Sort	$O(n^2)$
Insertion Sort	$O(n^2)$
Heap Sort	O(n logn)
Bellman –Ford Algorithm	O(V E)
Counting Sort to sort n elements (in the range : 0..k)	O(k + n)
Prim's algorithm	O(ElogV)
0/1 Knapsack using Brute Force	$O(2^n)$
Depth First Search using Adjacency List	O(V + E)

C. Answer the following question about Recurrence:
Using the master theorem, prove which algorithm is faster?
Justify your answer [5 marks]

A: T(n) = T (n /2) + 1
B: T(n) = 9 T(n/3) + n^2

Solution:-

A: T(n) = T (n /2) + 1 → or any other recurrences
Using the master theorem
$a = 1, b^d = 2^0$ → $a = b^d$ → $T(n) \in \Theta(n^d \log n)$
$T(n) \in O(\log n)$
B: $T(n) = 9\ T(n/3) + n^2$
$a = 9, b^d = 3^2$ → $a = b^d$ → $T(n) \in \Theta(n^d \log n)$
$T(n) \in O(n^2 \log n)$

➔ A is faster
// 2 marks for solving each recurrence relation 1 mark for the decision

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Question 2: To be answered in **(20) Minutes [ ] / 10 Marks**

You win a shopping voucher from Amazon. The rules of the voucher are that you can only take 8 (lbs) total of specific items (i.e. the Wight limit is 8). Each item has a weight and you can either take a whole item, or not take it at all.

Given the list of items below, Answer the following questions:

Item name	Item i	Value bi	Wight wi
Clock	1	15 $	1
CD player	2	10 $	5
Book	3	9 $	3
Bag	4	5 $	4

I. What does the **problem** of choosing items of maximal value with a specific weight call?
And cite one **technique** to solve this problem ? (2 marks)
…… Knapsack problem , Dynamic programming
……

II. Using the technique that you suggested above, Complete the matrix below :
(0.25 mark each) (4 marks)

Items	w[0]	w[1]	w[2]	w[3]	w[4]	w[5]	w[6]	w[7]	w[8]
0	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
1	**0**	**15**	**15**	**15**	**15**	**15**	**15**	**15**	**15**
2	**0**	**15**	**15**	**15**	**15**	**15**	**25**	**25**	**25**
3	**0**	**15**	**15**	**15**	**24**	**24**	**25**	**25**	**25**
4	**0**	**15**	**15**	**15**	**24**	**24**	**25**	**25**	**29**

III. Based on the filled array what **items** should you choose for maximum value less than or equal to 8 ?
What would be the **total value** in this case ? (3 marks)

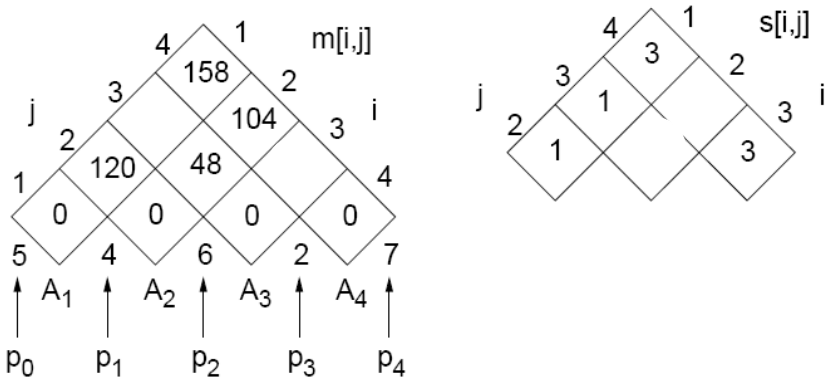
items 1,3, and 4 are selected
// 2 marks (0.25 for each number of item , 0.25 for not having any more
value is 29. // 1 mark

IV. What is the **running time** of the algorithm of this technique ? (1 mark)
…… O(n * W)
……

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Question 3 : To be answered in **(20) Minutes [ ] / 6 Marks**

For the problem of finding an optimal parenthesization of a matrix-chain product $<A_1,A_2,A_3,A_4>$ whose sequence of dimensions is (5, 4, 6, 2, 7) Answer the following questions: (1 mark for each point)



1. What is the optimal cost of parenthesizing this matrix chain $<A_1,A_2,A_3,A_4>$?
   // the answer is 158 (direct answer from the array m)
   .....
   .....

2. What is the optimal cost of parenthesizing the matrix chain $< A_3,A_4>$?

   // the answer is m[3,4]= 0+0+ 6 * 2 * 7 = 84
   .....
   .....

3. What is the range of **k** in the algorithm's formula of m [2,4]? // K could be 2 or k=3

   .....
   .....

4. What is the number that represented s[2,3]? // it should be 2

   .....
   .....

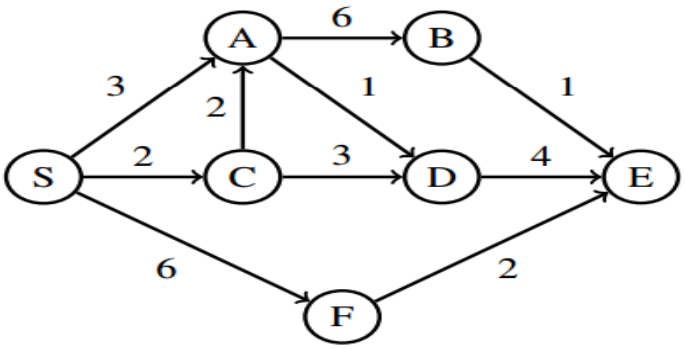
5. What is the optimal multiplication sequence for the matrix chain $<A_1,A_2,A_3>$?
   // $(A_1(A_2A_3))$
   .....
   .....

6. What is the optimal multiplication sequence for the matrix chain $\langle A_1, A_2, A_3, A_4 \rangle$:
// $((A_1(A_2A_3))A_4)$

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Question 4 : To be answered in **(20) Minutes [ ] / 11 Marks**

A. Use Dijkstra's Single Source Shortest Path algorithm on the following directed graph, starting at vertex S. *[4 marks]*



I. What is the order in which vertices get removed from the priority queue? (2 Marks)
……S, C, A, D, F, E, B.……..
……..

II. What is the shortest path from S to E? (2 Marks)
…… S, A, D, E ……..
……..

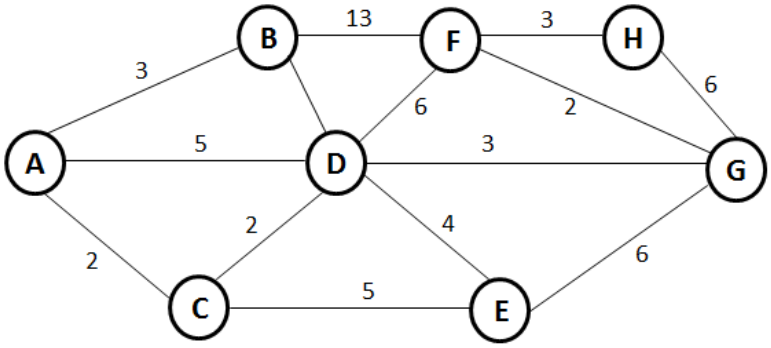
B. Consider a *complete undirected* graph with vertex set {a, b, c, d}. Entry W_{ij} in the matrix W below is the weight of the edge {i, j}. [2 marks]

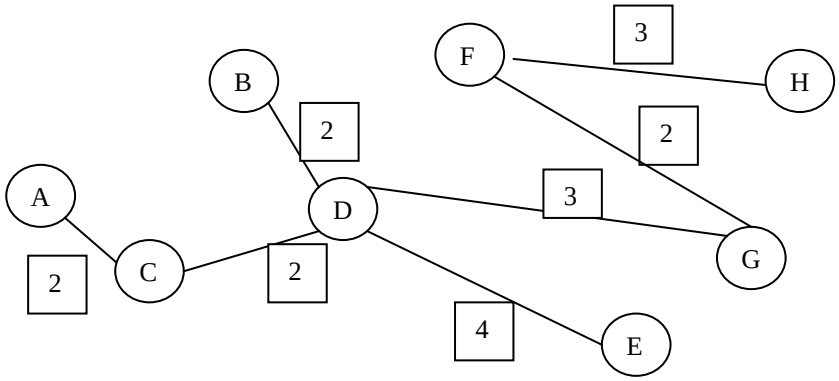
$$W = \begin{matrix} 0 & 1 & 8 & 1 \\ 1 & 0 & 12 & 4 \\ 8 & 12 & 0 & 7 \\ 1 & 4 & 7 & 0 \end{matrix}$$

What is the minimum possible weight of a spanning tree T in this graph such that **vertex a** is a leaf node in the tree T?

……
……

C. Find The Minimum Spanning Tree of the following graph using Kruskal's algorithm.
Show the steps and the resulting Minimum Spanning Tree and give its weight. *[5 marks]*





The Minimum spanning Tree weight is …**18.**...

End of questions

Good Luck